Business Statistics

Visualizing Numerical Variables

Datasets used in this presentation and accompanying MS Excel models are simulated by artificial intelligence. While they are realistic, they are not from actual companies or pre-existing data sets.

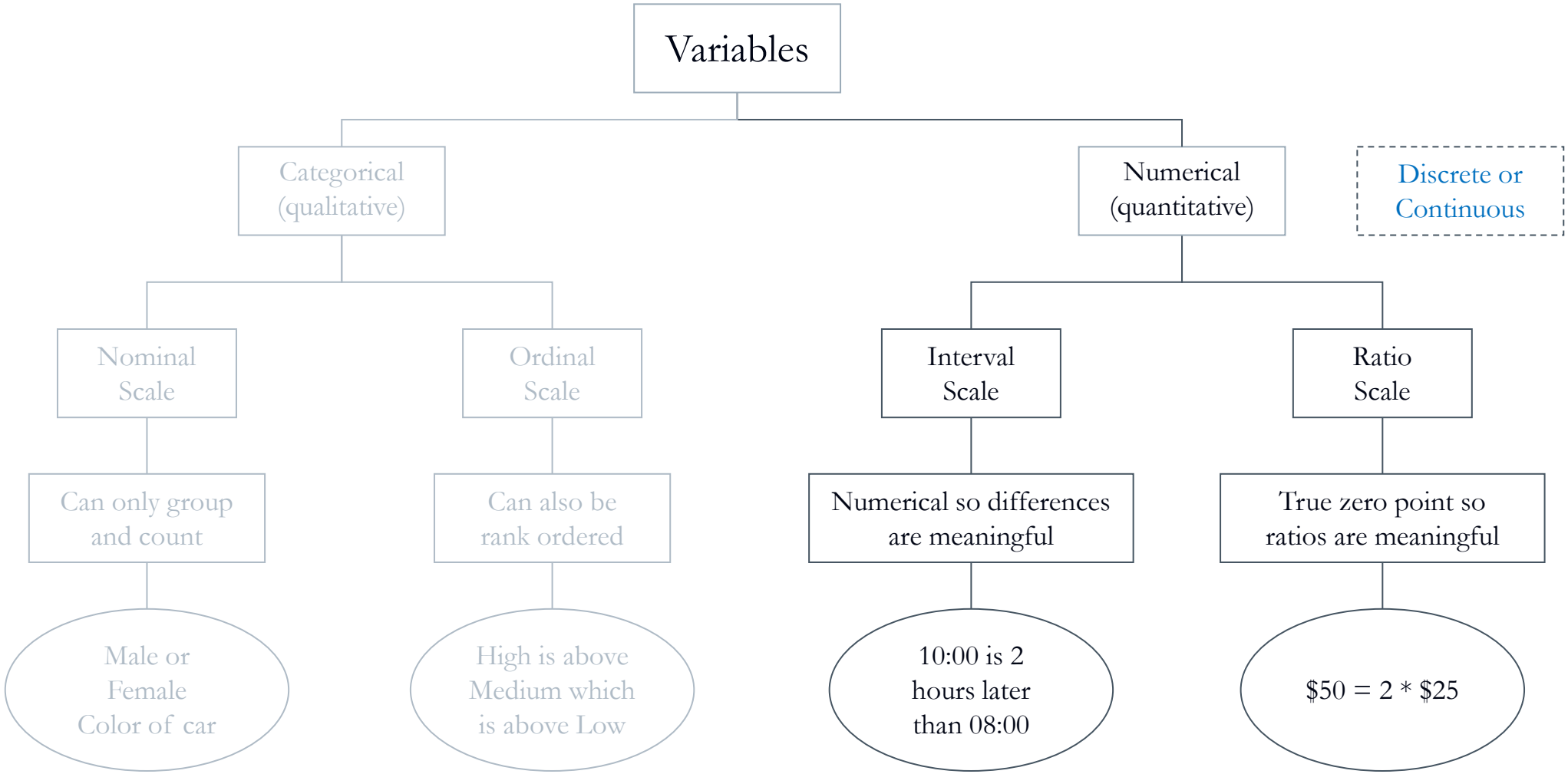
Visualizing Numerical Variables
Overview

- Interval and ratio variable types

- Visualizing one Numerical variable
  - Preparing the data: ordering, frequency distributions
  - Histogram

- Visualizing two Numerical variables
  - Ordered pairs
  - Scatterplot

Numerical data
Variable types are based on their measurement scale



Visualizing one Numerical variable

Preparing the Data: Frequency Distribution

Visualizing Numerical Variables
Preparing data: frequency distribution

- Raw data:
  - List of 30 product purchase amounts
  - Ratio data
- Order the observations
- Create bins and frequencies
  1. Establish bins (classes, intervals)
  2. Get frequencies by bin

Raw data

PurchaseID	Purchase Amounts ($)
P001	60
P002	15
P003	135
P004	20
P005	22
P006	190
P007	27
P008	30
P009	55
…	…
P028	275
P029	95
P030	8

Visualizing Numerical Variables
Preparing data: order observations

- Excel has formulas for ordered statistics (median, percentiles)
- Still, ordered data gives a big picture and pattern
  - Only 7 of the 30 purchases are greater than $100

Raw data

PurchaseID	Purchase Amounts ($)
P001	60
P002	15
P003	135
P004	20
P005	22
P006	190
P007	27
P008	30
P009	55
…	…
P028	275
P029	95
P030	8

Ordered data

PurchaseID	Purchase Amounts ($)
P030	8
P018	12
P002	15
P024	18
P004	20
P005	22
P026	25
P007	27
P008	30
…	…
P027	225
P028	275
P022	340

Visualizing Numerical Variables
Preparing data by creating bins

- Counts of numerical data require that the values of multiple observations be grouped
- Grouping numerical variable observations transforms the numerical variable into a categorical variable for visualization
- The different names for these ranges of numerical values are synonymous, with slight distinctions in general:
  - **Bins** are used with computers/software (Excel)
    - Bins only have an upper, **inclusive** bin limit
  - **Classes** and **intervals** generally are for statistics
    - Classes have both a lower **inclusive** and upper **exclusive** boundary
- The practitioner must decide how many bins to make and their ranges

Visualizing Numerical Variables
Preparing data: number of bins and size

- The number/size of bins should give information about the data
  - Generally, there should be between 5 and 15 bins
  - Too few bins can hide details
  - Too many bins will not summarize the data
- The bins must be **MECE (Mutually Exclusive Collectively Exhaustive)**
  - Mutually Exclusive: no overlap, each observation belongs in only one bin
  - Collectively Exhaustive: no exclusions, every observation belongs in a bin
- Determining the number and size of bins can be an iterative process
  - The ordered data may suggest a starting point for number of bins

Visualizing Numerical Variables
Preparing data: Start by choosing number of bins

- Start by selecting the number of bins and calculating bin width:
  - Observations range from 8 to 340
  - Guess 10 bins
  - Bin width: (340 – 8) / 10 = 33.2
  - Round to 35

$$\text{Bin Width} = \frac{(\text{Max value} - \text{Min value})}{\text{Number of Bins}}$$

- First 2 bins' observations are near 10
- All others have 2 or less observations
- This is not a good summary and is not an informative binning scheme

Bin upper limit	Bin	Frequency
35	0 < amt <= 35	11
70	35 < amt <= 70	10
105	70 < amt <= 105	2
140	105 < amt <= 140	2
175	140 < amt <= 175	1
210	175 < amt <= 210	1
245	210 < amt <= 245	1
280	245 < amt <= 280	1
315	280 < amt <= 315	0
	315 < amt	1

While bins only include an inclusive upper bin limit, we show both bin limits for exposition.

Visualizing Numerical Variables
Preparing data: Or start by choosing bin width

- Alternatively start by selecting bin width:
  - Observations range from 8 to 340
  - Guess a bin width of 50, giving 7 bins

- 17 observations in first bin
- 6 observations in second bin
- 5 bins with 2 or less observations
- Context determines if this is an informative binning scheme
- Do we need granularity of those observations in the first bin?

Bin upper limit	Bin	Frequency
50	0 < amt <= 50	17
100	50 < amt <= 100	6
150	100 < amt <= 150	2
200	150 < amt <= 200	2
250	200 < amt <= 250	1
300	250 < amt <= 300	1
	300 < amt	1

Visualizing Numerical Variables
Preparing data: modifying the overflow limit

- The overflow limit is the lower limit of the uppermost bin
- It captures all observations that are greater than that limit
- Lowering that limit reduces bins and groups outliers

Original binning by 50 gave 7 bins with 3 bins of 1 observation

Bin upper limit	Bin	Frequency
50	0 < amt <= 50	17
100	50 < amt <= 100	6
150	100 < amt <= 150	2
200	150 < amt <= 200	2
250	200 < amt <= 250	1
300	250 < amt <= 300	1
	300 < amt	1

Reducing overflow to 200 gives 5 bins with 3 in the overflow

Bin upper limit	Bin	Frequency
50	0 < amt <= 50	17
100	50 < amt <= 100	6
150	100 < amt <= 150	2
200	150 < amt <= 200	2
	200 < amt	3

Visualizing Numerical Variables
Preparing data: relative frequencies

- Adding the relative and percent frequencies helps to understand the relative distribution of the observations across the bins
- This can also be done during the bin selection process

Bin upper limit	Bin	Frequency	Relative Frequency	Percent Frequency	Cumulative Frequency
50	0 < amt <= 50	17	0.57	57%	57%
100	50 < amt <= 100	6	0.20	20%	77%
150	100 < amt <= 150	2	0.07	7%	83%
200	150 < amt <= 200	2	0.07	7%	90%
	200 < amt	3	0.10	10%	100%
	Total	30	1.00	100%	100%

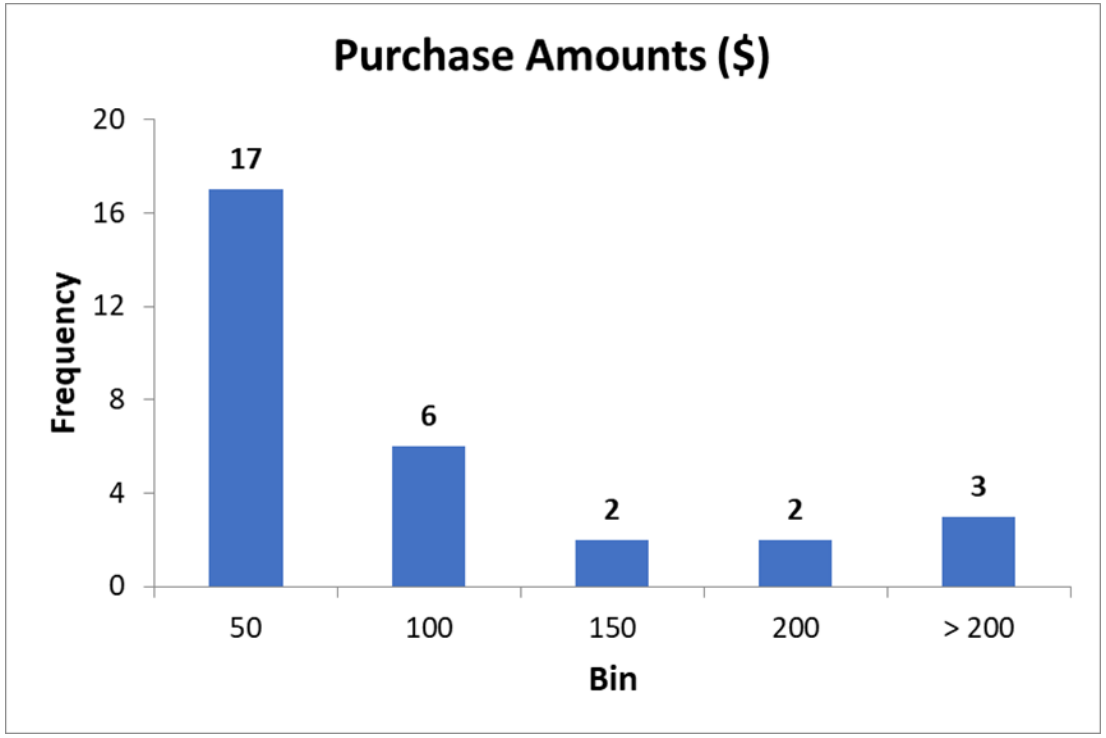
Visualizing one numerical variable

Histogram

Visualizing One Numerical Variable Histograms

- Histograms are the frequency counts of numerical variable values by bin (class, interval)

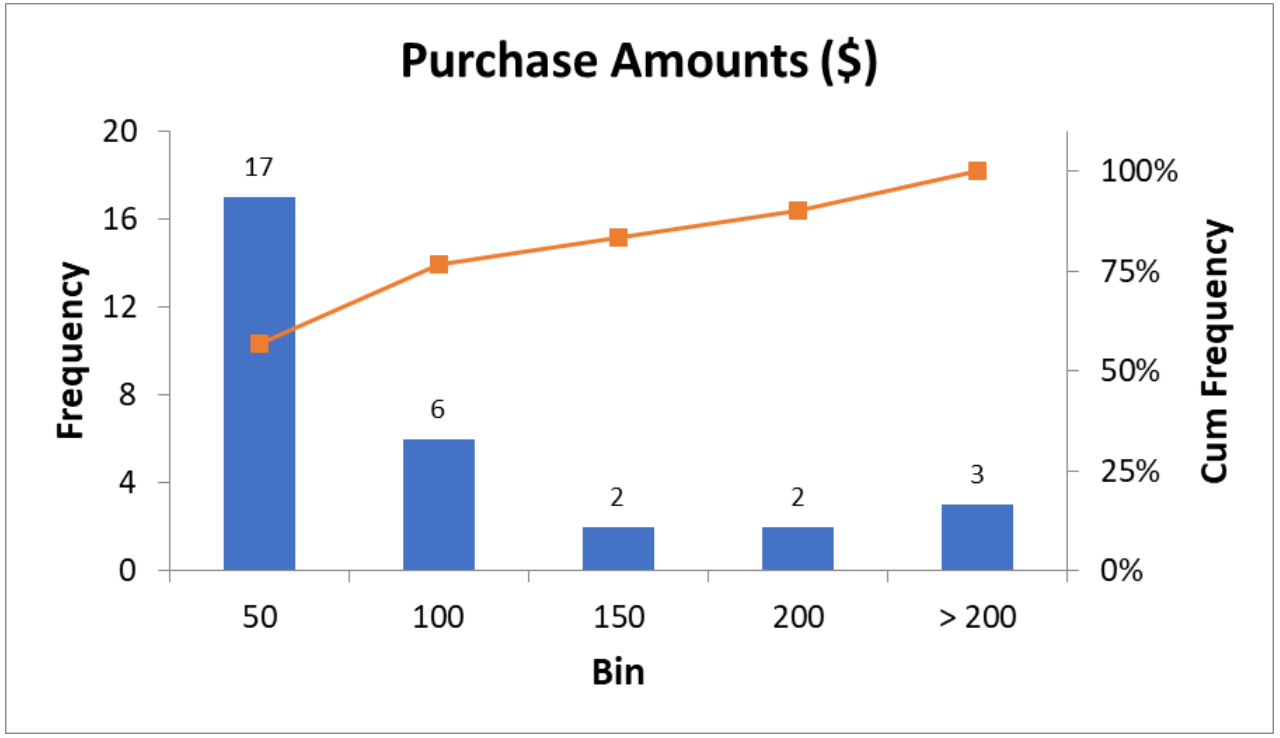
Bin upper limit	Bin	Frequency
50	0 < amt <= 50	17
100	50 < amt <= 100	6
150	100 < amt <= 150	2
200	150 < amt <= 200	2
	200 < amt	3



Visualizing One Numerical Variable
Histograms

- Histograms are the frequency counts of numerical variable values by bin (class, interval)
- The cumulative frequency is also often helpful

Bin upper limit	Bin	Frequency	Cumulative Frequency
50	0 < amt <= 50	17	57%
100	50 < amt <= 100	6	77%
150	100 < amt <= 150	2	83%
200	150 < amt <= 200	2	90%
	200 < amt	3	100%



Visualizing two numerical variables

Scatter plot

Visualizing Two Numerical Variables
Seeing the relationship

- The purpose of displaying two numerical variables on the same chart is to see if a relationship exists

- Raw data:
  - 30 months of advertising spend and revenue
  - Ratio data

- Ensure units are appropriate and consistent

Month	Advertising($K)	Revenue($K)
1	8	54
2	8	49
3	10	62
4	10	57
5	12	65
6	12	52
7	15	60
…	…	…
28	45	115
29	45	100
30	50	130

Visualizing Two Numerical Variables
Scatter plot

- A scatter plot displays the data as ordered pairs of variables

$$(x, y)$$

- Ordered pairs will be the pairs of values for each month

$$(\mathbf{advertising_1}, \mathbf{revenue1})$$

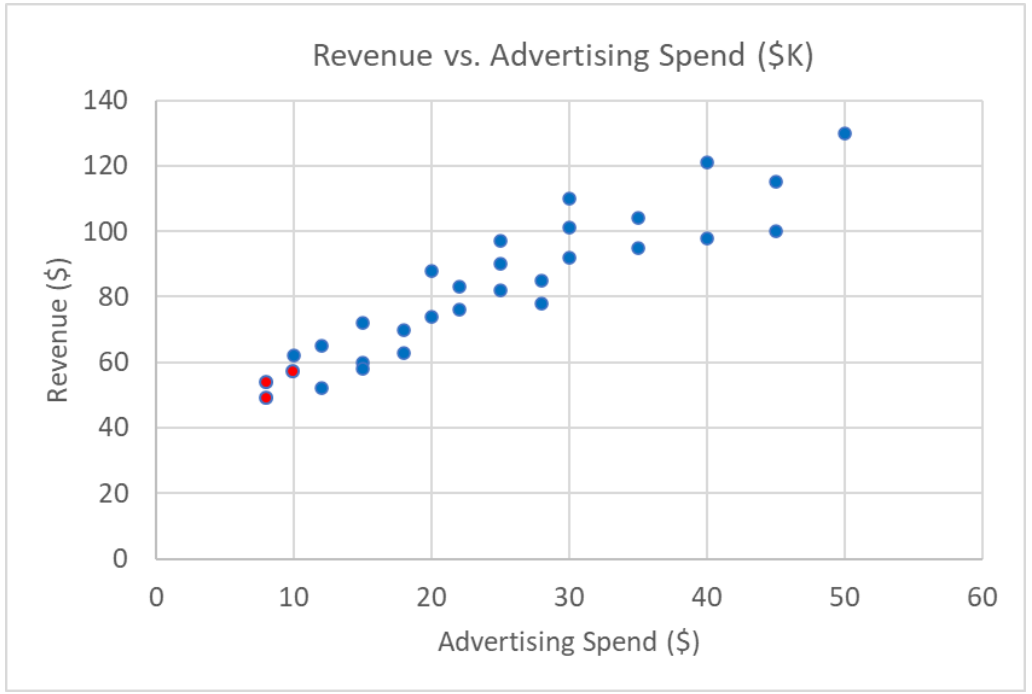
- Month 1 $(\mathbf{8}, \mathbf{54})$
- Month 2 $(\mathbf{8}, \mathbf{49})$
- Month 3 $(\mathbf{10}, \mathbf{62})$

Month	Advertising($K)	Revenue($K)
1	8	54
2	8	49
3	10	62
4	10	57
5	12	65
6	12	52
7	15	60
…	…	…
28	45	115
29	45	100
30	50	130

Visualizing Two Numerical Variables
Scatter plot shows if a relationship exists

- Horizontal axis: advertising
- Vertical axis: revenue

- Observations:
  - Month 1 $(\mathbf{8}, \mathbf{54})$
  - Month 2 $(\mathbf{8}, \mathbf{49})$
  - Month 3 $(\mathbf{10}, \mathbf{62})$



- Is there a relationship between advertising spend and revenue?
- Generally, as the advertising increases, the revenue increases

Visualizing Two Numerical Variables
Scatter plot with a trendline

- For this data a trendline clearly shows the relationship
- Caution: depending on the data, relying on a trendline can lead to completely incorrect conclusions

